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Heterogeneous Effects and the Interpretation of OLS Coefficient Movements

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Abstract

Applied researchers commonly interpret coefficient movements across OLS and difference-in-differences specifications with varying controls as evidence of bias reduction or improved identification. This note shows that such interpretations generally fail under heterogeneous treatment effects. In this case, each specification estimates a differently weighted average treatment effect, where the weights depend on the set of included controls. Consequently, adding or removing controls can change the estimand even when the controls are irrelevant for the outcome conditional on treatment. The note further derives an augmented version of the canonical omitted variables bias formula that remains valid under heterogeneity by explicitly accounting for induced changes in weighting.

Keywords: ordinary least squares, heterogeneous treatment effects, model specification and testing
JEL Codes: C31, C51

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1 Introduction

Empirical research in economics commonly reports sequences of OLS and difference-in-differences (DiD) specifications with progressively richer sets of controls. A standard practice is to interpret coefficient movements across these specifications as evidence about omitted variable bias or improved identification.

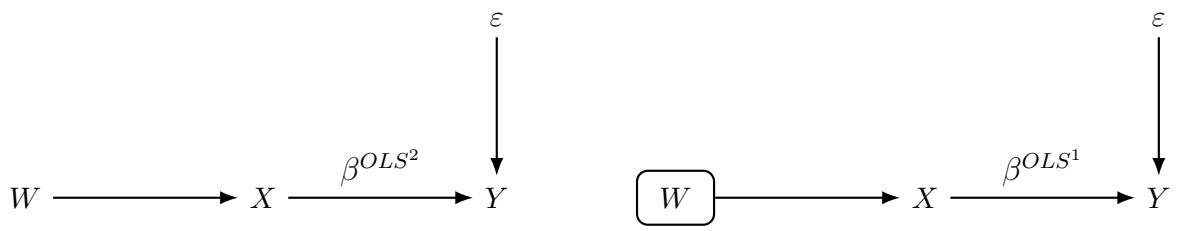
This note argues that such interpretations are generally invalid once treatment effects are heterogeneous. In that case, linear estimators do not recover a single structural parameter but instead estimate specification-dependent weighted averages of unit-specific treatment effects. Crucially, the weights depend on the variation in the regressor used for estimation, which changes whenever controls or fixed effects are added.

As a consequence, even control variables that are conditionally irrelevant for the outcome can change the estimand by altering the residual variation in the treatment variable. Coefficient movements across specifications may therefore reflect changes in weighting rather than changes in bias or identification quality.

This mechanism is closely related to the interpretation of instrumental variables (IV) estimators as local average treatment effects (LATE). However, the implication for OLS and DiD is less widely emphasized: these estimators are also local objects under heterogeneity, with specification-dependent weighting schemes.

Figure 1: (Not) controlling for "irrelevant" W makes a difference, $\beta^{OLS^1} \neq \beta^{OLS^2}$

(a) Without control W (b) With control W



Notes: Both panels depict the same causal structure in which W affects X but has no direct effect on Y. In the presence of heterogeneous effects of X on Y, the OLS coefficient depends on how variation in X is weighted across groups. Controlling for W changes the residual variation in X and therefore the implicit weights, implying $\beta^{OLS^1} \neq \beta^{OLS^2}$.

Figure 1 illustrates the key mechanism. Even when W has no direct causal effect on Y, conditioning on W changes the residual variation in X. Under heterogeneous treatment effects, this implies a change in the implicit weighting of group-specific effects, since OLS aggregates treatment effects using the variance of the identifying variation in the regressor. As a result, specifications with and without W generally estimate different convex combinations of heterogeneous effects, even if W is irrelevant in the sense of satisfying standard exogeneity conditions.

This observation has direct implications for the interpretation of coefficient movements in applied work. In particular, it shows that differences between nested specifications cannot, in general, be attributed solely to changes in omitted variable bias. Instead, they may also reflect changes in the underlying estimand induced by reweighting across heterogeneous strata. Building on this mechanism, the note derives an augmented version of the canonical omitted variable bias formula that explicitly accounts for changes in the weighting structure across specifications with heterogeneous treatment effects.

The note contributes to three strands of the literature. First, it relates to the large literature on OLS weighting under heterogeneous treatment effects, including Angrist and Krueger (1999) and Angrist and Pischke (2008) and more recent work on regression and matching estimands (Chattopadhyay and Zubizarreta, 2022, Mogstad and Torgovitsky, 2024). Second, it connects to the literature on coefficient stability and omitted variable bias diagnostics in applied work (Altonji et al., 2005, Oster, 2019, Pei et al., 2019). Third, it complements the IV literature emphasizing that estimators identify weighted local effects rather than global parameters.

The main point of the paper is not to introduce new econometric tools, but to highlight a simple implication of existing results that is often overlooked in applied work: in the presence of heterogeneous effects, specification changes in linear models change not only bias properties but also the estimand itself.

2 OLS with heterogeneous effects

2.1 Setup

Consider the model

$$Y_i = \beta_g X_i + \varepsilon_i, \quad g \in \{1, \ldots, G\}, \tag{1}$$

where treatment effects vary across latent strata g, which are unobserved by the researcher. Let $\pi_g = \Pr(g)$ denote group shares.

2.2 Pooled OLS estimand

The pooled OLS coefficient can be written as:

$$\beta^{OLS} = \frac{\sum_{g=1}^{G} \pi_g \beta_g \mathrm{Var}(X \mid g)}{\sum_{g=1}^{G} \pi_g \mathrm{Var}(X \mid g)}. \tag{2}$$

Defining weights,

$$w_g = \frac{\pi_g \mathrm{Var}(X \mid g)}{\sum_{h=1}^{G} \pi_h \mathrm{Var}(X \mid h)}, \tag{3}$$

we obtain:

$$\beta^{OLS} = \sum_{g=1}^{G} w_g \beta_g, \quad \sum_g w_g = 1. \tag{4}$$

OLS therefore aggregates heterogeneous effects using weights determined jointly by group size and within-group variation in the regressor.

The key implication is that OLS does not estimate a structural average effect unless $\mathrm{Var}(X \mid g)$ is constant across groups. Otherwise, OLS places greater weight on groups with more variation in the identifying regressor.[1]

2.3 Residualisation and weighting

Now consider:

$$Y_i = \beta X_i + \boldsymbol{W}'\gamma + \varepsilon_i, \tag{5}$$

with $E[\boldsymbol{W}\varepsilon] = 0$ and $Cov(X, \boldsymbol{W}) \neq 0$.

[1] For a numerical example, see Appendix A.1.

By Frisch–Waugh–Lovell:

$$\beta^{PM} = \frac{Cov(\tilde{X}, Y)}{Var(\tilde{X})}, \tag{6}$$

where $\tilde{X}$ is the residual from projecting X on $\boldsymbol{W}$.

Under heterogeneous effects:

$$\beta^{PM} = \sum_{g=1}^{G} w_g^{PM} \beta_g, \tag{7}$$

with

$$w_g^{PM} = \frac{\pi_g \mathrm{Var}(\tilde{X} \mid g)}{\sum_{h=1}^{G} \pi_h \mathrm{Var}(\tilde{X} \mid h)}. \tag{8}$$

Proposition 1. OLS with controls identifies the average treatment effect if and only if $\mathrm{Var}(\tilde{X} \mid g)$ is constant across groups.

Proof. The estimator is a convex combination of g with weights proportional to $\mathrm{Var}(\tilde{X} \mid g)$. These weights reduce to population shares only if the variance term is constant across strata. Otherwise, OLS reweights treatment effects across groups. □

Thus, including controls changes the estimand whenever they differentially affect the residual variation of X across groups. Appendix A.2 provides a numerical example.

3 The augmented formula for omitted variable bias

Let β^S denote the short regression and β^L the long regression. Then:

$$\beta^L - \beta^S = \underbrace{\sum_g (w_g^L - w_g^S)\beta_g}_{\text{heterogeneity reweighting}} + \underbrace{\gamma \cdot \frac{\mathrm{Cov}(W_i, X_i)}{\mathrm{Var}(X_i)}}_{\text{classical OVB}}, \tag{9}$$

where the first term captures changes in weighting induced by controls.

This decomposition shows that coefficient movements are not diagnostic of omitted variable bias alone but include changes of the estimand.

4 Discussion and Conclusion

With heterogeneous treatment effects, OLS estimates are not structural parameters but specification-dependent weighted averages. These weights depend on the distribution of identifying variation in the regressor, which changes whenever control variables are included.

As a consequence, coefficient movements across specifications cannot be interpreted solely through omitted variable bias logic. They may instead reflect changes in the implicit aggregation of heterogeneous effects.

This result applies equally to difference-in-differences and other linear panel estimators. Analogous to the multiple OLS case, adding time-varying controls in a difference-in-differences setting affects the residualised variation in the regressor across groups, thus changing their relative weighting and the estimand.

Experimental designs are not affected due to orthogonal assignment of treatment. Here, the group-specific variance can (only) change due to changes in group size, as recently pointed out by Słoczyński (2022).

A comparison with IV estimates is also informative, which are generically understood to target different weighted subpopulations defined by the instrument. Going back to Figure 1: Notice how W qualifies as instrument, where the regression of X on W is the first stage. In contrast, through Frish-Waugh-Lovell, OLS that conditions on W provides an estimate for the population estimand of everyone except the complies of this particular instrument. This is because the IV first stage regression and the auxiliary regression from Frish-Waugh-Lovell are one and the same specification, so their estimands speak to the opposing sides of the same coin.[2]

Overall, model comparisons across specifications are not informative about causal robustness unless effect homogeneity or stability of identifying variation across groups can be justified. This also carries implications for machine learning approaches that choose controls or models based on statistics of overall fit, such as the R^2 or Mean-

[2]To illustrate this property, Appendix A.3 provides a numerical example that replicates a pattern common to mny IV papers through changes of the estimand, and without any violation of classical endogeneity assumptions: estimates first become smaller when more controls are added, but the IV is larger.

Squared-Error. Such models usually assume identical estimands across specifications, which is not warranted with effect heterogeneity.

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Appendix

A.1 Numerical example to illustrate the weighting in simple OLS

The Stata code available here generates a single dataset with $N = 1000$, for a data generating process where Y depends only on X and a normally distributed error term only.[3] To abstract from weighting due to group size, there exist two equally sized strata ($g = 1$ for obs. 1-500, $g = 2$ for obs. 501-1000). The variance in X is not constant across strata, we have $VAR(X_g|1) < VAR(X_g|2)$, so the regressor values are independent but not identically distributed. The outcome is defined as $Y = \beta_g X + \epsilon$ and treatment effects are heterogeneous with $\beta_1 = 1$ and $\beta_2 = 5$. Note that in this example it is the strata with the larger β_g that also has the higher variance in X_g. This means OLS will put excessive weight onto the strata with the higher effect so that $\beta^P > \beta^{AE}$.

Table A.1 shows in column (1) the OLS estimate only considering the observations for the first strata, in column (2) only for the second strata. For completeness, in column (3) we have $\hat{\beta}^{AE}$ which uses only the sample size to weight groups. Last but not least, column (4) shows the pooled OLS estimate $\hat{\beta}^P$.

The estimates in columns (1) and (2), where treatment effects are homogeneous, behave as expected. The "hand-constructed" average effect $\hat{\beta}^{AE}$ in column (3) is close to three, too. However, $\hat{\beta}^P$ is far away from this average effect of a one-unit change in X on Y. This is because observations from the second group (with larger effects) are over-weighted due to their larger variance.

Notice that in this numerical example, we set the $VAR(X)|1 = \frac{1}{3}VAR(X)|2$. Such a difference in treatment variation can occur for a binary treatment for example if half of the individuals who would really benefit from a treatment (strata 2) take the treatment, but only about ten percent of individuals from strata 1 (who benefit much less) do so, see in particular Słoczyński (2022). Such differences in treatment variation across strata,

[3] Of course, this exercise can be repeated for distributions of estimates. Here, for simplicity and illustration, only a single sample realisation of the data-generating process is discussed.

Table A.1: Estimates with heterogeneous effects and heteroskedastic strata that are positively related

	(1) $\hat{\beta}_1$	(2) $\hat{\beta}_2$	(3) $\hat{\beta}^{AE}$	(4) $\hat{\beta}^{P}$
X	0.957***	4.979***	2.975	4.004***
	(0.0457)	(0.0253)	-	(0.0831)
Constant	0.0964*	0.0332	-	0.138
	(0.0455)	(0.0440)	-	(0.0815)
N	500	500	1000	1000

Notes: This table presents simple OLS estimates of a generated dataset where treatment effects are heterogeneous across two strata and the variance in the regressor is not constant across groups. The exact Stata code for replication can be found here. Hereroskedasticity-robust standard errors are in parenthesis. *** denotes significance at the 1%-level, * denotes significance at the 10%-level.

violating the assumption of identically distributed regressors, are not implausible.

In practice, the applied researcher rarely knows the sizes or identities of strata, so that only β^P can be estimated. This should not be interpreted as an estimate of an average population parameter without the explicit assumption of homogeneous treatment effects or heterogeneous effects with constant variance in X across (unknown) strata.

A.2 Numerical example: "Irrelevant" controls move the estimate

Table A.2 reports estimates of a single realisation of a data-generating process where continuous Y depends on continuous X and on a normally distributed disturbance term only (Stata code available here). The researcher has access to an additional continuous variable W that can (not perfectly) correlate with X but, to repeat for clarity, does not have direct effects on Y. Columns (1) to (4) show the OLS estimates for two samples that have true βs equal to one and five respectively. In these samples, W represents what is routinely referred to as "irrelevant control". Adding W indeed has little effect on the estimates, which are close to their true (population) parameters.

In practice, substrata, i.e. groups that have different effect sizes, are rarely known,

Table A.2: Irrelevant controls become relevant in pooled OLS

	Sample 1		Sample 2		Pooled Sample	
	(1)	(2)	(3)	(4)	(5)	(6)
X	0.957***	0.957***	4.964***	5.017***	3.000***	2.466***
	(0.0457)	(0.0457)	(0.0438)	(0.0979)	(0.113)	(0.0893)
W		-0.00800		-0.0545		1.061***
		(0.0426)		(0.0909)		(0.0732)
constant	0.0964*	0.0965*	0.0332	0.0320	0.103	0.0960
	(0.0455)	(0.0455)	(0.0440)	(0.0440)	(0.0691)	(0.0617)
N	500	500	500	500	1000	1000

Notes: This table presents OLS estimates of a generated dataset where treatment effects are heterogeneous across two strata where the variance in the regressor constant across groups. The exact Stata code for replication can be found here. Hereroskedasticity-robust standard errors are in parenthesis. *** denotes significance at the 1%-level, * denotes significance at the 10%-level.

and instead the applied researcher estimates a single β coefficient in a pooled sample. In our example, the average effect in the population of a one-unit change in X equals three. Column (5) shows the OLS estimate over the pooled sample that is very close to this population parameter.

And here comes the (potential) surprise: adding the previously irrelevant control W to this pooled sample in column (6) changes the estimate. The estimated effect of a one-unit change in X on Y is now 2.466, highly significant at conventional levels and different to the previous estimate and from the average population parameter (which is known and equals three).

A.3 OLS and IV estimates across specifications with exogeneity but reweighting

The weighting of multivariate OLS and in IV estimation are directly related. A particular feature of multivariate OLS is that adding more controls to capture potential endogeneity can weigh the estimates away from "complier" populations, which might have larger true treatment effects compared to never-takers, for example.

Such underlying DGPs can generate commonly observed sequences of decreasing

multivariate-OLS estimates when more controls are added, and larger IV estimates, just through weighting and without violations of the independence assumption. The Stata code available here provides a numerical example of such a pattern, which is shown by Table A.3.

Table A.3: Smaller OLS estimates and larger IV without endogeneity

	(1) $\hat{\beta}^{P}$	(2) $\hat{\beta}^{PM}$	(3) $\hat{\beta}^{PM}$	(4) $\hat{\beta}^{IV}$
X	3.000***	2.466***	0.955***	4.914***
	(0.113)	(0.0893)	(0.0457)	(0.191)
W		1.061***		
		(0.0732)		
V			4.010***	
			(0.0634)	
Constant	0.103	0.0960	0.0647*	0.187*
	(0.0691)	(0.0617)	(0.0316)	(0.0906)
Controls included		✓	✓	
N	1000	1000	1000	1000

Notes: This table presents OLS estimates of a generated dataset where treatment effects are heterogeneous across two strata where the variance in the regressor constant across groups. W is a valid instruments and therefore "irrelevant" control. V equals X for observations of the second strata and is zero otherwise. The exact Stata code for replication can be found here. Hereroskedasticity-robust standard errors are in parenthesis. *** denotes significance at the 1%-level, * denotes significance at the 10%-level.